



# ALI ARIF

## Software Engineer

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Lahore, Pakistan

## SKILLS

### Back-End:

Python

Django

Django DRF & ORM

### Database:

PostgreSQL

MS SQL

Amazon RDS

### Cloud & DevOps:

AWS EC2

S3

SES

CloudFront

RDS

Docker

Route 53

CloudWatch

### AI-Powered Tools & Frameworks:

Pinecone Vector DB

Semantic Search

Embeddings

LangChain

Chatgpt

Claude

Cursor

### Core Development Skills:

Web App

SAAS

Web API

Web Crawling

LLM Integration

RAG

Async Tasks

Stripe

JWT

AI Chatbot Development

Role-Based Access Control

Authentication

OAuth2

Social Logins

Logic Building

### Tools & Technologies:

Postman

Sendgrid

Git

VS Code

Redis

Dbeaver

Celery

Facebook SDK

Beautiful Soup (BS4)

ReportLab (PDF Generation)

Google OAuth

## SUMMARY

*Backend-Focused Software Engineer with 5+ years of experience in designing and developing scalable systems using Python, Django, and RESTful APIs. Expertise in database design, authentication, async processing, automation and building AI-powered backend architectures. Experienced with AWS cloud services and microservice-based architectures, delivering robust production-ready backend solutions.*

## EXPERIENCE

### Backend & AI Software Engineer | Highline Residential

February 2026 – Present

Remote, New York, USA

- Designed scalable Django REST APIs and optimized PostgreSQL/PostGIS search systems for a large-scale MLS-backed real estate platform.
- Reduced listing search latency by 88% through PostgreSQL/PostGIS query optimization, indexing strategies, and search architecture redesign.
- Developed AI-powered media generation pipelines using OpenAI, AWS, Replicate, Fireworks AI, and BrowserShots.
- Built amenities extraction and validation pipelines that combine AI workflows with deterministic processing.
- Collaborated on scalable backend architecture, API reliability, geospatial systems, and AI-assisted operational workflows for a live US real estate platform.

### AI Instructor (GenAI) | ICMA Digital Academy

December 2025 – Present

Remote, United Kingdom

- Deliver the "Gen-AI & Business Productivity" program covering AI fundamentals, prompt engineering, AI applications, & chatbot development.
- Conduct live lectures, hands-on labs, assessments, & project mentoring using Gemini, NotebookLM, Google AI Studio, & IBM watsonx.

### Software Engineer (Freelance/ Contract) | Self Employed

July 2025 – Present

Remote, Pakistan

- Delivered production SaaS applications using React, Django, AWS, & AI-powered workflows.
- Built multi-tenant architectures, RBAC systems, authentication, & API integrations.
- Implemented Celery, Redis, RabbitMQ, & webhook-driven automation pipelines.
- Developed chatbot integrations across Meta platforms, Zapier, & third-party business systems.

### Software Engineer | Public Sector Organization (NESCOM)

January 2022 – June 2025

Islamabad, Pakistan

- Engineered secure Django REST APIs with RBAC, JWT/OAuth2, & Celery-based processing.
- Developed RAG-powered chatbot platforms using OpenAI, Pinecone, embeddings, & semantic search.
- Built automated training pipelines for web, PDF, DOCX, & structured data ingestion.

# SKILLS

## Architecture:

MVC

MVT

RESTful APIs

Feature-based (Frontend)

Microservices

## Design Patterns & Principles:

Singleton

Facade

Repository

SOLID

## Practices & Methodologies:

Agile Development (Jira)

Problem-Solving & Debugging

Security Best Practices

Performance Optimization

# DISTINCTION

 Gold Medal  
BSEE

# AWARDS

 Dean's Honour List  
MSEE

 Graduate Fellowship  
MSEE

 Dr. M. Nawaz Medal  
Best Final Year Project in  
BSEE

 Dean's Honour List  
BSEE

 Excellence Award  
Co-curricular activities in  
UCP

 1st Runner-up  
Engineering Project Ex-  
hibition in SOFTEC'18,  
NUCES, LHR

# LANGUAGES

English: **Business Fluent**

Urdu: **Native**

# REFERENCES

Engr. M. Uzair Azhar

Senior Manager, Public Sector  
Organization

@uzairazhar@outlook.com

- Developed Stripe-based billing & analytics platforms for SaaS applications.
- Optimized PostgreSQL/MS SQL databases & AWS-based deployments for scalable production workloads.

## System Software Engineer | Rapidev Pvt. Ltd. (NSTP, NUST)

 January 2021 – December 2021

 Islamabad, Pakistan

- Designed, tested, and validated system-level models using STK, MATLAB, CST, and SystemVue.
- Collaborated on scalable system architectures, verification, and performance optimization.

## Research Associate | Information Technology University (ITU)

 September 2016 – December 2020

 Lahore, Pakistan

- Led computational and biomedical engineering research projects and supervised graduate/fellowship year students.

# EDUCATION

## M.Sc. in Electrical Engn. (MSEE) | Information Technology University (ITU)

 October 2016 – August 2018

 Lahore, Pakistan

- GPA: 3.60/4.00 (Grade A+)

## B.Sc. in Electrical Engineering (BSEE) | University of Central Punjab (UCP)

 October 2012 – June 2016

 Lahore, Pakistan

- GPA: 3.95/4.00 (Grade A+)

# PROJECTS

## Real Estate & Community Discovery Platform | Casamere

Contributed to a large-scale MLS-backed real estate and community discovery platform featuring geospatial search, AI-powered media pipelines, and operational tooling.

- Engineered scalable Django/PostGIS search APIs with map-based filtering and community discovery workflows.
- Improved listing search performance from 90s to 11s through query optimization, indexing strategies, and database redesign.
- Developed AI-powered media generation and amenities extraction pipelines using OpenAI, Fireworks AI, Replicate, and AWS services.

**Skills Used:** Python, Django, PostgreSQL, PostGIS, OpenAI, AWS, Redis, Celery

## Social Media Management SaaS | Socifiy

Architected and delivered a production-grade SaaS platform for social media management, scheduling, collaboration, and workflow automation.

- Built multi-tenant backend services with JWT authentication, RBAC, Google OAuth, and Meta integrations.
- Developed React dashboards, Celery-based scheduling workflows, and scalable REST APIs.
- Deployed cloud-native infrastructure on AWS using EC2, S3, CloudFront, RDS, Docker, and CI/CD workflows.

**Engr. M. Hamza Adeem**  
Software Engineer, Ericsson  
@ mhamza.adeem@gmail.com

**Dr. Ali Nasir**  
Asst Prof, King Fahd University  
of Petroleum & Minerals  
@ ali.nasir@kfupm.edu.sa

**Dr. M. Qasim Mehmood**  
Chairperson, EED, Information  
Technology University  
@ qasim.mehmood@itu.edu.pk

**Skills Used:** Django, React, PostgreSQL, Redis, AWS, Celery, Meta APIs

## Omnichannel AI Chatbot Integrations (Meta + Slack) | [SmartChatAI](#)

Built a multi-tenant messaging and automation platform integrating AI chatbots with Meta and Slack communication channels.

- Implemented secure webhook ingestion, encrypted token management, and real-time message routing.
- Developed idempotent processing, retry mechanisms, monitoring, and reliability-focused backend systems.
- Designed scalable chatbot architectures supporting asynchronous workflows and conversation management.

**Skills Used:** Python, Django, Webhooks, Redis, Celery, Meta APIs, Slack APIs

## RAG-Based AI Chatbot Engine & Training System

Developed a scalable AI chatbot platform with vector search, automated training pipelines, and context-aware conversational agents.

- Built LangChain-equivalent architecture using OpenAI, Pinecone, embeddings, and semantic retrieval.
- Developed automated multi-source ingestion pipelines supporting websites, PDFs, DOCX, XLSX, and Q&A datasets.
- Implemented conversational memory, prompt orchestration, analytics, and AI performance monitoring.

**Skills Used:** OpenAI, Pinecone, Django, PostgreSQL, Celery, Redis, BeautifulSoup, Django ORM, REST APIs

## Billing & Business Intelligence Platform

Delivered a subscription billing and analytics platform integrating Stripe for enterprise SaaS operations.

- Built subscription lifecycle management including plans, trials, add-ons, renewals, invoices, and webhooks.
- Developed payment automation, lead capture, reporting, and customer management workflows.
- Implemented notification systems and operational automation using SES, Celery, and Redis.

**Skills Used:** Stripe API, Django, PostgreSQL, AWS SES, Celery, Redis

## CERTIFICATES

🌟 Google AI Specilization | [Google](#)

🌟 Gen-AI & Business Productivity Specialization | [IBM](#) | [Cambridge University](#)

🌟 Version Control; Programming with JavaScript; Introduction to Front-End Development | [META](#)

🌟 AWSOME DAY ONLINE CONFERENCE 2025; AWS Cloud Technical Essentials | [AWS](#)

🌟 Machine Learning and Data Science Bootcamp | [Udemy](#)

🌟 Path Certificates: Building a To-Do App with React; Fundamental Coding Prep with Python; Unlocking Data Insights with PostgreSQL | [CodeSignal](#)

🌟 Python; Pandas; Intro to Machine Learning; Intermediate Machine Learning | [Kaggle](#)

## PUBLICATIONS

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- A Novel Cesaro Fractal EBG-Based Sensing Platform for Dielectric Characterization of Liquids, in IEEE Transactions on Antennas and Propagation, May 2021.
- A Low-Cost Multiple CSRR-Based Microwave Sensor for Contactless Dielectric Characterization of Liquids, in IEEE Sensors Journal, Oct, 2020.
- A Compact, Low-Profile Fractal Sensor for Wearable On-Body WBAN Applications, in IEEE Antennas and Wireless Propagation Letters, May 2019.